CHAPTER TWO: WEB PROJECT MANAGEMENT



Learning Objectives:

By the end of this chapter the learner shall be able to;

i. Explain the process of project management ii. Explain online project management iii. Estimate the cost and the size on an online project

2.1 Introduction to Project management

Project management is the discipline of planning, organizing, securing and managing resources to bring about the successful completion of specific project goals and objectives. A *project* is a temporary endeavor, having a defined beginning and end (usually constrained by date, funding and deliverables), undertaken to meet unique goals and objectives, usually to bring about beneficial change or additional value. The temporary nature of projects stands in contrast to business (or operations), which are repetitive, permanent or semi-permanent functional work to produce products or services.

The primary challenge of project management is to achieve all of the project goals and objectives while honoring the preconceived project constraints. Typical constraints are scope, time, and budget. The secondary, and more ambitious, challenge is to optimize the allocation and integration of inputs necessary to meet pre-defined objectives.

The following are the major project management processes/steps: -

1. *Initiation* - The initiation processes determine the nature and scope of the project. If this stage is not performed well, it is unlikely that the project will be successful in meeting the business needs. The initiation stage should include a plan that encompasses the following areas:

   ○ Analyzing the business needs/requirements in measurable goals ○ Reviewing of the current operations ○ Financial analysis of the costs and benefits including a budget ○ Stakeholder analysis, including users, and support personnel for the project ○ Project charter including costs, tasks, deliverables, and schedule

2. *Planning and design* - After the initiation stage, the project is planned to an appropriate level of detail. The main purpose is to plan time, cost and resources adequately to estimate the work needed and to effectively manage risk during project execution. Project planning generally consists of: - ○ Determining how to plan ○ Developing the scope statement ○ selecting the planning team ○ identifying deliverables and creating the work breakdown structure

   ○ identifying the activities needed to complete those deliverables and networking the activities in their logical sequence ○ estimating the resource requirements for the activities ○ estimating time and cost for activities ○ developing the schedule ○ developing the budget ○ risk planning ○ gaining formal approval to begin work

3. *Executing* - Executing consists of the processes used to complete the work defined in the project management plan to accomplish the project's requirements. Execution process involves coordinating people and resources, as well as integrating and performing the activities of the project in accordance with the project management plan. The deliverables are produced as outputs from the processes performed as defined in the project management plan.

4. *Monitoring and controlling* - Monitoring and controlling consists of those processes performed to observe project execution so that potential problems can be identified in a timely manner and corrective action can be taken, when necessary, to control the execution of the project. The key benefit is that project performance is observed and measured regularly to identify variances from the project management plan. Monitoring and Controlling includes:

   - Measuring the ongoing project activities ('where we are')
   - Monitoring the project variables (cost, effort, scope, etc.) against the project management plan and the project performance baseline (where we should be)
   - Identify corrective actions to address issues and risks properly (How can we get on track again)
   - Influencing the factors that could circumvent integrated change control so only approved changes are implemented

5. *Closing* - Closing includes the formal acceptance of the project and the ending. Administrative activities include the archiving of the files and documenting lessons learned. This phase consists of:

    - Project close: Finalize all activities across all of the process groups to formally close the project or a project phase

    - Contract closure: Complete and settle each contract (including the resolution of any open items) and close each contract applicable to the project or project phase.

2.2 Online project management systems

Online project management systems allows for project management tasks to be done over the Internet. Online project management systems allow any authorized person connected to the network to check the status of the project. The system centralizes all the information in a location that can be accessed easily by others.

Basic Online Project Management Functions

The purpose of an online project management system is to organize and simplify projects from start to finish. The following are the basic functions that almost every online project management system has:

- *Organizational systems* - keep track of files and documents. Most allow users to customize the filing system. The goal of the system is to make it easy to find any single piece of information in a project, no matter how complex that project may be.

- *Document management* - tracks document modifications, including the identity of the person who changed the file. Many systems archive older copies of files rather than delete them.

- *Scheduling* - is another key feature for most systems. Whether it's a centralized calendar with to-do lists or an automated schedule of e-mails to keep everyone informed and on task, scheduling plays a big role in project management. Many systems include streamlined features that let managers keep track of who's doing what and the project's progress as a whole.

- *Time tracking* - with this function, software keeps track of how much time people spend working on each part of a project. Many companies bill customers based on how much time was spent on a particular project. Others might pay employees at an hourly rate. Time tracking applications let the company keep tabs on how many hours each person in the project has worked on various phases of the project.

- *Documentation* – this involves keeping a detailed history of projects, which can help protect a company from liability. For example, let's say a construction company gets a request to build a house. The client makes a change to the house's design early in the project. The construction company makes a note of the client's requests and documents it using an online project management system. Later, when construction is complete, the client expresses dissatisfaction with the way the house looks due to the changes in design. With the documentation feature, the construction company has a record that can help it prove that the changes were at the client's request.

- *Issue tracking* - lets users create a file to report any problems or issues that crop up during a project. These applications have a feature that e-mails the appropriate parties so that the issue can be resolved as quickly as possible. Once someone fixes the problem, he or she can note it in the issue report. The system then files the report as part of the documentation function.

- *Communication systems* - might include a message board, instant messaging program, chat room or some other communication forum. Including a message system within the overall system makes it easier for people involved in the project to pass information along.

2.3 Examples of web projects

- *New web side design and development.*

- *Web Site Redesign:* if it has been more than 3 years since your last design, it is definitely time to at least take a serious look at your web site

- *Adding Functionalities to Your Web Site:* time to start using ecommerce to sell products or process donations, add an interactive calendar, add social net working, etc.

- *Search Engine Optimization:* if you‘d like to capture more search traffic for your products or services, it simply won‘t happen without conscious SEO

- *Web Site or Internet Marketing*: if you build it, they will come is not a sound business practice, you need a plan to promote awareness of your site

- *Email Campaigns:* email has enjoyed a revival and when a campaign is properly constructed, the results can be quite extraordinary

- *Developing a mobile version.*

2.4 Choosing a Web development firm

In order to find a Web design firm best suited for your project, your organization should be aware of the following:

- The intended goal of the project
- The intended audience for the Web site
- The anticipated budget for the Web site development
- How the project will fit into a larger marketing/corporate scheme
- The individuals/departments within your organization responsible for the project

Five steps to follow when choosing a Web design company.

- Self Analysis - define your organization's requirements. That said, the better you can define your Web development and marketing goals the better it is – for both you and the potential Web development firm.
- Generate a list – gather a list of potential Web design from referrals, analysis of other sites you admire and from list provided by a consultant.
- Create a shortlist - From the list of potential Web design firms, your organization's goal should now be to create a shortlist of 3-5 firms that are best suited for your project. In order to determine which Web design firms are most appropriate for your project, do the following:

  ○ Examine their portfolio ○ Technical competence and experience ○ Analyze the process ○ Determine what other products/services the Web design company can offer Get proposals - Once you have selected your short-list of top Web development firms, request that they send you a proposal. Ask that their proposal include an overview of your requirements and their proposed solution. This will help to determine which Web design firm understands your requirements best. Also ask that each Web design firm include a description of their development process and a price breakdown for the various aspects of the Web site project.

- Evaluate and select a partner - Start off by examining each proposal individually. After your organization has reviewed all proposals, compare them with each other. Once you have made a decision, contact the top Web development firm and let them know the good news.

2.5 Budget Planning for the Web site development project

First, make sure the scope is defined for the Web presence. A documented Web operations strategy will help you define budget parameters so when you make a request, you can clearly state that you're seeking funding for "X" Web property.

Next, state the goals for the Web property and the planned strategies and tactics for meeting those goals. This process will allow you to identify quantifiable success metrics and provide rationale for funding.

At this point, you can begin to assign costs to each line item. Make sure you take into consideration the costs for hardware, software, design, implementation and maintenance. Other cost considerations are training, communications and change management.

Web goals should include specific projects (i.e. a site re-design) as well as day-to-day operations. Your budget should account for all the things that are required for maintaining a quality Web presence. This should include daily upkeep of design and editorial content, ongoing usability and analytics, R&D, and maintenance of Web tools and infrastructure.

Depending on how your organization budgets, you may have a stand-alone Web budget or you may need to collaborate with other departments to make sure they're accounting for the Web project in their budgets.

A detailed Web plan for upcoming projects and day-to-day operations will help everyone understand priorities and costs for the Web presence. Share the plan with your Web stakeholders to ensure buy-in and support for the budget.

The following are the main reasons why web projects get over budget: -

Frequently changing requirements - rigid requirements make it difficult to react to the changes that inevitably occur as knowledge and environments evolve. Requirements that have been omitted are generally picked up late in the process, by which time they are awkward and costly to implement.

- Too many stakeholders having a say in the web development process
- Not enough budget or time being allocated

How to ensure that web design and development projects are delivered in budget: -

- Take a collaborative approach to Web project development - The approach would involve regular meetings with all stakeholders, where working software is tested and enough quality assurance is carried out.
- By using a combination of web development, project management and software engineering tools
- Making sure that requirements were collected exhaustively
- Allocating enough time and budget to the web project

2.6 Cost estimation for web projects

Software cost models and effort estimates help project managers allocate resources, control costs and schedule and improve current practices, leading to projects finished on time and within budget. In the context of Web development, these issues are also crucial and very challenging given that Web projects have short schedules and very fluidic scope. Web projects cost estimation is the process through which the web developer decides how much the development process of the website will cost the organization.

For Web development, cost is difficult to estimate because:

- There is no standard to sizing Web applications. Each can be created using diverse technologies such as several varieties of Java (Java, servlets, Enterprise java Beans, applets, and Java Server Pages), HTML, JavaScript, XML, XSL, and so on.
- Web project‘s primary goal is to bring quality applications to market as quickly as possible, varying from a few weeks
- People involved in Web development are represented by less experienced programmers, users as developers, graphic designers and new hires straight from university
- Processes employed are in general heuristic, although some organizations are starting to look into the use of agile methods
- Web projects have short schedules and a fluidic scope
- Typical project size is small, using 3 to 7 team members

Several techniques for cost and effort estimation have been proposed over the last 30 years in software engineering, falling into two general categories: -

1. *Expert judgement (EJ)* – EJ has been widely used. However, the means of deriving an estimate are not explicit and therefore not repeatable. Expert opinion, although always difficult to quantify, can be an effective estimating tool on its own or as an adjusting factor for algorithmic models

2. *Algorithmic models (AM)* –attempts to represent the relationship between effort and one or more project characteristics. The main ―cost driver‖ used in such a model is usually taken to be some notion of software size (e.g. the number of lines of source code, number of pages, and number of links). Algorithmic models need calibration or adjustment to local circumstances.

2.7 How to measure the size of the website

Cowderoy (1998; 2000) recommends several size measures for cost estimation and risk assessment of Web application development projects. Measures have been organised into the following categories: -

- ***Web-site design.*** Creation and organisation of Web pages. **Length measures** provided are Web pages (number of Web pages), home pages (number of major entry points to the Web application/site), leaf nodes (number of leaf Web pages, i.e., pages that have no siblings), hidden nodes (number of pages excluded from the main navigation buttons), depth (number of nodes on the second level that have siblings); **complexity measures** are interconnectivity (number of URLs that link to other pages in the same application/site), external hyperlinks (number of unique URLs in the Web application/site); **functionality measures** are actions (number of independent actions provided from a Web page by use of Javascript, Active X and Java applets).
- **Technical authoring.** Collection of text and related hyperlinks, and text structuring sections etc. **Length measures** proposed are paragraph count (number of paragraphs in a Web page, Web application, or external document, i.e., PDF document), word count (number of words in a Web page or document); **complexity measures** are navigational structures (number of different structures in a Web page or document).
- ***Data entry.*** Creation and maintenance of databases used by a Web application. **Length measures** proposed are key record types (number of different tables accessed by the application/site), key entries (number of keys in a database table), entry fields (number of table attributes manipulated by the application/site), secure fields (number of fields that contain safety-critical or financially secure information).
- ***Graphics design.*** Creation of 2D images using graphics design tools. **Length measures** are delivered images (number of unique images used by the Web application/site), image size (width * height), image composites (number of layers from which the final image was created),

language versions (the number of versions of an image that must be produced to accommodate different languages or different cultural priorities).

- ***3D and VR(Virtual Reality) design*.** Creation of 3D objects and environments with which the user can interact. **Length measures** are 3d objects (number of files including one or more 3D objects used in the Web application/site) and virtual worlds (number of files including one or more virtual worlds used in the Web application/site).
- ***Sound engineering*.** Adaptation of existing audio tracks for use at Web applications. **Length measures** are audio files (number of unique audio files used in the Web application/site), duration (summed duration of all sequences within an audio file), audio sequences (number of sequences within the audio file), tracks (number of layers within an audio sequence), audio edits (number of edits applied to groups of frames within a sequence, e.g., fades) and imported images (number of separate graphics images imported into an interactive audio file).
- ***Animation and video editing*.** Building movies from existing images, video, text and sound. **Length measures** are movies (number of movie files used in a Web application, Web page, or referenced by other movies at this Web application or Web page), duration of a movie, and story composites (for each scene, count the number of layers from which the final image was created), frame rate (number of frames per second as defined in the delivered file).
- ***Programming*.** Creation of server-side software and client-side programs that replace browsers. Classifies measures into three categories – **Specification, Design and Program**. The **Specification length measures** proposed are user commands (number of different menu commands, buttons, etc that the user can access from the application‘s interface), database files (number of entities and relationships in the database schema) and classes definitions (object-oriented classes resulting from the analysis of use cases). The **Design length measures** are any design object-oriented measures where no one in particular is specified. The **Design functionality measures** are Function Points or Full Function Points. Finally, **Program length measures** are lines of source code.

Chapter Review Questions

1. Define project management.
2. Outline the major steps of project management
3. What are the major functions of online project management
4. What are the main reasons why web projects are over budget?

References

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Suggested Further Reading

1. Niederst J., Learning Web design, Shroff Publishers
2. Peter, K., Web design, Tools and Techniques, Peach Pit press
3. http://www.w3schools.com